Subject: Computer Networks

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Assignment No.: 1

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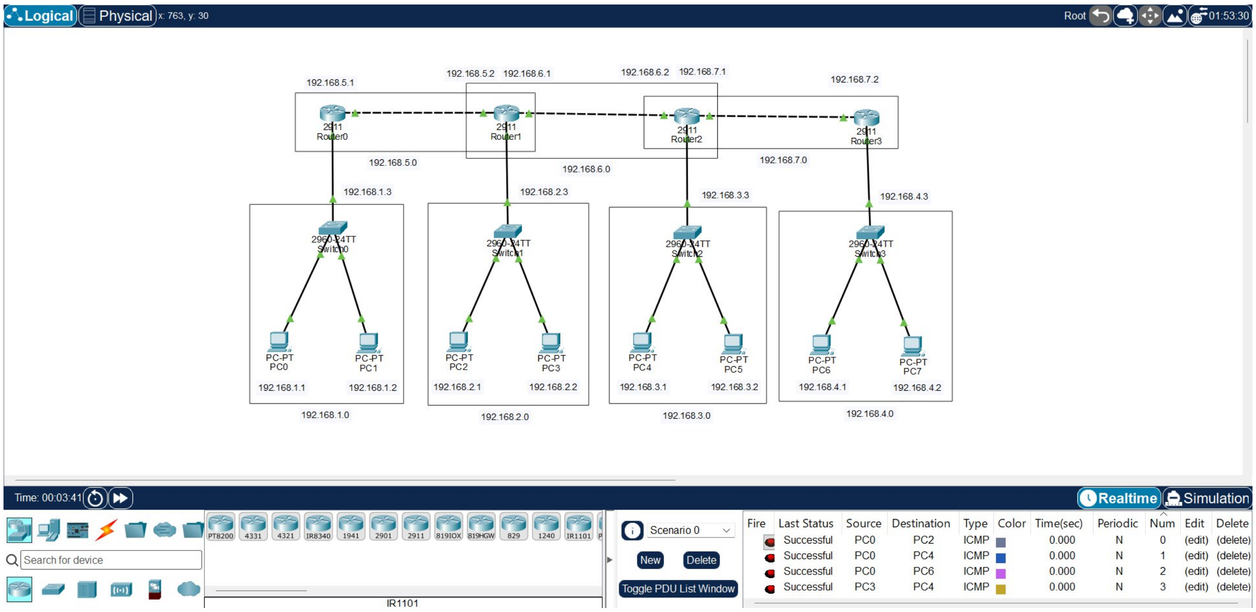
Assignment-1

Q1. Develop Static Routing with 4 routers, 4 Switches & 8 PC's. Make sure to highlight the Networks with their network ID's. IP Addresses should be written in text format as well. (6 M)

Ans –



The network is designed using four routers, four switches, and eight PCs to demonstrate static routing. Each LAN is assigned a unique network ID, and separate networks are used for the connections between routers. IP addresses are assigned to all PCs and router interfaces and are displayed in the topology.

Q2. What is OSI model? What are the different layers of OSI model? Write the functionality, Data format and Protocols used at each layer. (3 M)

Ans – The OSI (Open Systems Interconnection) Model is a conceptual framework developed by the International Organization for Standardization (ISO) to standardize communication between different computer systems over a network. It provides a common structure for understanding how data is transmitted from one device to another.

The OSI model divides the network communication process into seven independent layers, where each layer performs a specific function and provides services to the layer above it. The layers work together to ensure reliable and efficient communication between the sender and receiver. Each layer handles a particular part of the communication process and follows specific protocols.

The seven layers of the OSI model, from top to bottom, are:

Application à Presentation à Session à Transport à Network à Data Link à Physical

The OSI model helps in designing, implementing, troubleshooting, and understanding computer networks by dividing the complex communication process into manageable layers.

Name	Functionality	Data Format	Protocols
Application	It provides network services to end-user applications.	Data	HTTP, FTP, SMTP, DNS
Presentation	It handles data translation, encryption/decryption and compression.	Data	SSL/TLS, JPEG, ASCII
Session	This layer establishes, manages and terminates communication sessions.	Data	NetBIOS, RPC
Transport	It provides end-to-end delivery, error control, flow control and segmentation.	Segment/Datagram	TCP, UDP
Network	It provides logical addressing and routing of data between networks.	Packet	IP, ICMP, OSPF, RIP
Data Link	It provides framing, MAC addressing and error detection for local delivery.	Frame	Ethernet, PPP, Wi-Fi
Physical	It transmits raw bits through physical media such as cables and wireless signals.	Bits	Ethernet, USB, Bluetooth

Q3. What are the different Error Detection methods? Explain each with the help of proper examples. (3M)

Ans – Error detection is the process of identifying errors that occur in data during transmission from the sender to the receiver. Errors may occur due to noise, interference, or signal distortion during transmission. Error detection methods add some extra bits to the original data so that the receiver can determine whether the received data is correct or corrupted. Some common error detection methods are Parity Check, Checksum, and Cyclic Redundancy Check (CRC).

i) Parity Check – Parity Check is a simple error detection technique in which an additional bit called a parity bit is added to the transmitted data. The parity bit is selected so that the total number of 1s in the transmitted data becomes either even (even parity) or odd (odd parity). At the receiver, the number of 1s is checked. If the expected parity is not maintained, an error is detected.

Example:
Data = 1011001 (four 1s)

For even parity, parity bit = 0

Transmitted data = 10110010

If one bit changes during transmission, the number of 1s may become odd, and the receiver can detect that an error has occurred.

ii) Checksum – Checksum is an error detection technique in which the data is divided into fixed-size blocks. These blocks are added using binary addition, and the 1's complement of the final sum is calculated. This complement is called the checksum and is transmitted along with the original data. At the receiver, the data blocks and checksum are added again. If the result is not the expected value, an error is detected.

Example:
1010 + 0101 = 1111
Checksum (1’s complement of sum) = 0000
Transmitted data: 1010 0101 0000

iii) Cyclic Redundancy Check (CRC) – Cyclic Redundancy Check is a powerful error detection technique based on binary modulo-2 division. It is widely used in computer networks and data communication because it can detect many types of transmission errors. In CRC, the sender divides the data, after appending zeros, by a predetermined generator using XOR operation. The remainder obtained from this division is called the CRC. This remainder is appended to the original data and transmitted. At the receiver, the complete received data is divided by the same generator. A zero remainder indicates that no error has been detected, whereas a non-zero remainder indicates an error.

Example:

Data = 110101

Generator = 1011

CRC Remainder = 111

Transmitted data: 110101111